

MAKAUT

RESEARCH METHODOLOGY

A Last Minute Preparation Book

DEBKANTA MAITY

DEBKANTA STUDY ZONE [DSZ]

1. Discuss the significance of accuracy, precision, and reliability in research.

1) Accuracy :

Accuracy refers to the closeness of a measured or observed value to the true or accepted value.

Significance :

- Ensures that the results are correct and reflect the actual situation.
- High accuracy reduces systematic errors (bias).
- It is essential for drawing valid conclusions and making proper decisions.
- Increases the scientific value and credibility of the research.

Example :

If the true value of length is 10.00 cm and we measure 10.02 cm, the result is accurate (close to true value).

2) Precision :

Precision refers to the closeness of repeated measurements to each other, i.e., consistency of results.

Significance :

- Ensures reproducibility and consistency in measurements.
- High precision means low random errors.
- Helps in detecting small differences or changes.
- Essential for comparing results across time, places, or experiments.

Example :

If the length is measured as 10.02 cm, 10.03 cm, 10.01 cm, 10.02 cm, these values are precise (close to each other) but may not be accurate.

3) Reliability :

Reliability refers to the consistency and dependability of a measurement instrument or research procedure over time and under different conditions.

Significance :

- Provides trust and confidence in the research findings.
- Reliable instruments and methods produce stable and repeatable results.
- Ensures that the research can be replicated by other researchers.
- Increases the generalizability of the results.

Example :

If the same experiment done today, tomorrow and by another researcher gives similar results, the research is reliable.

* Relationship among Accuracy, Precision and Reliability :

Accuracy	→ Closeness to true value (correctness)
Precision	→ Closeness to each other (consistency)
Reliability	→ Consistency of instrument/process over time and conditions (dependability)

* Summary :

For high quality research, we need results that are accurate, precise and reliable.
⇒ They form the foundation of valid, credible and trustworthy research.

Accuracy gives correctness, Precision gives consistency and Reliability gives trustworthiness to research.

2. Discuss the importance of method validation in research.

Definition :

Method validation is the process of demonstrating that a research method or analytical procedure is suitable for its intended purpose and produces reliable, accurate and consistent results.

Importance of Method Validation :

- ① Ensures **Accuracy and Reliability** : Validated methods produce results that are accurate and reliable, reducing errors and uncertainty.
- ② Increases **Credibility of Research** : Validated methods increase the trust and confidence of researchers, peer reviewers and the scientific community in the results.
- ③ Ensures **Reproducibility** : Validation ensures that the method can be reproduced by other researchers under similar conditions, which is a key principle of scientific research.
- ④ **Detects and Minimizes Errors** : It helps to identify potential sources of error or bias in the method and minimizes their impact on the results.
- ⑤ Determines **Suitability for Intended Purpose** : It confirms that the method is appropriate for the specific type of sample, measurement range, and research objectives.
- ⑥ Helps in **Regulatory Compliance** : In many fields (e.g., pharmaceuticals, biotechnology, food safety), method validation is required by regulatory bodies to ensure quality and safety.
- ⑦ Supports **Data Interpretation** : Validated methods provide confidence that the results truly reflect the actual situation, helping in correct interpretation and decision making.
- ⑧ **Optimizes Resources** : By validating a method, unnecessary repetition, time and cost are saved, as the method is proven efficient and effective.
- ⑨ Facilitates **Comparison of Results** : When the same validated method is used across different studies or laboratories, results can be compared and combined meaningfully.
- ⑩ Strengthens **Scientific Conclusions** : It ensures that the conclusions drawn from research are based on sound, proven and trustworthy methods.

In summary :

Method validation is essential to ensure that research methods are accurate, reliable, reproducible and fit for purpose, which ultimately leads to high quality research and scientific progress.

Q.3 Differentiate between accuracy and precision.

3

Both accuracy and precision are essential in research, but they measure different aspects of data quality.

ACCURACY

• Definition:

Accuracy refers to the closeness of a measurement to the true or accepted value.

- It indicates how correct a result is.

• Mathematical Expression:

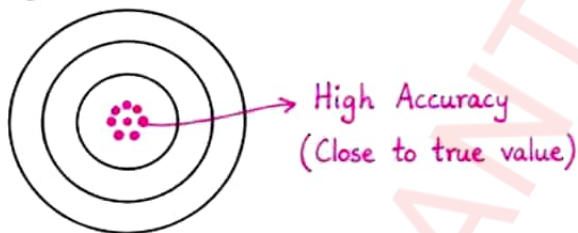
Let $\bar{x}$ = measured value
 x_t = true value

$$\text{Absolute Accuracy} = |\bar{x} - x_t|$$

• Characteristics:

- High accuracy means the result is very close to the true value.
- Accuracy is affected by systematic errors (bias).

• Diagram (Target Representation):



Key Differences:

- Accuracy is about closeness to the true value.
- Affects trueness of results.
- Caused by systematic errors.
- Measured by $|\bar{x} - x_t|$.
- **Example:** A weighing scale giving exact weight.

PRECISION

• Definition:

Precision refers to the closeness of repeated measurements to each other, i.e., the reproducibility or consistency of results.

- It indicates how consistent a result is.

• Mathematical Expression:

Let $x_1, x_2, \dots, x_n$ be repeated measurements.

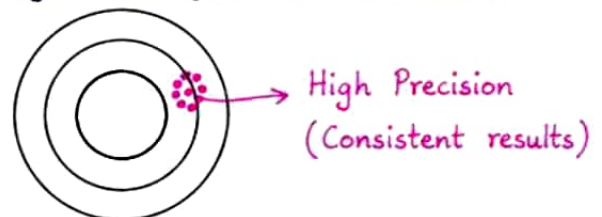
$$\text{Mean } (\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Precision (Spread)} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

• Characteristics:

- High precision means results are tightly clustered.
- Precision is affected by random errors.

• Diagram (Target Representation):



- Precision is about closeness among repeated results.
- Affects consistency of results.
- Caused by random errors.
- Measured by $\frac{1}{n} \sum (x_i - \bar{x})^2$.
- **Example:** A scale giving same reading repeatedly.

Summary:

Accuracy tells "how close we are to truth".
Precision tells "how close we are to each other".

Ideal Case:



High Accuracy
and
High Precision.

Q.4 Discuss advantages and limitations of observation techniques.

4

Observation is a data collection method in which the researcher watches and records events, behaviors, or phenomena as they occur naturally.

Advantages:

- ① Real and Natural Data
→ Data is collected as events occur in natural setting.
- ② First-hand Information
→ Researcher gets direct and original information without relying on others.
- ③ Useful for Sensitive Topics
→ Helps in studying behaviors or events that people may not report truthfully.
- ④ No Response Bias
→ Since there are no questions asked, it avoids bias due to personal opinions or false reports.
- ⑤ Helpful in Exploratory Research
→ Useful when little is known about the problem; provides insights and new hypotheses.
- ⑥ Cross-checking of Data
→ Can be used to validate information obtained from other sources.
- ⑦ Captures Non-verbal Behavior
→ Allows recording of gestures, expressions, body language, etc.
- ⑧ Suitable for Various Fields
→ Applicable in psychology, sociology, education, marketing, healthcare, management, etc.

Limitations:

- ① Time Consuming
→ Observation takes a lot of time, especially in natural settings.
- ② Observer Bias
→ Personal beliefs, feelings or expectations of the observer may influence the observation.
- ③ Limited Generalizability
→ Results may not be generalized to a larger population.
- ④ Reactivity (Hawthorne Effect)
→ Subjects may change their behavior because they know they are being observed.
- ⑤ Selective Attention
→ Observer can record only what he/she sees; important events may be missed.
- ⑥ Difficult to Quantify
→ Some behaviors or events are hard to measure in numerical terms.
- ⑦ Ethical Issues
→ Involves privacy, consent, and possible invasion of personal space.
- ⑧ Not Suitable for All Situations
→ Cannot be used where events are rare, private, or not visible.

At a Glance

Advantages

- Provides real and natural data.
- Direct and first-hand information.
- No response bias.
- Useful for sensitive and exploratory studies.
- Helps in cross-checking and validation.
- Captures non-verbal behavior.
- Applicable in various fields.

Limitations

- Time consuming and costly.
- Affected by observer bias.
- Limited generalizability.
- Subjects may react to being observed.
- Observer has selective attention.
- Hard to quantify some behaviors.
- Raises ethical and privacy concerns.
- Not suitable for rare or private events.

★ Key Takeaway:

Observation technique is powerful because it collects real and natural data, but it requires careful planning and awareness of its limitations to ensure valid and reliable results.

Q5. Discuss various methods of data collection.

Data collection is the systematic process of gathering information relevant to the research problem.

1. PRIMARY METHODS (First-hand Data)

- ① **Observation Method** : Data is collected by observing events, behaviors, or phenomena in their natural setting.

Key Points :

- Can be participant or non-participant.
- Useful for studying actual behavior.
- Requires careful and unbiased observation.

- ② **Interview Method** : Data is collected by asking questions and recording the responses.

Key Points :

- Can be structured, semi-structured, or unstructured.
- Allows in-depth understanding.
- Time-consuming and interviewer bias may occur.

- ③ **Questionnaire Method** : A set of written questions is given to respondents to fill in their answers.

Key Points :

- Cost-effective and can reach many people.
- Responses depend on how questions are framed.
- Lower response rate may be a limitation.

- ④ **Schedule Method** : Similar to a questionnaire, but the investigator asks questions and fills in the answers.

Key Points :

- Useful for illiterate or less-educated respondents.
- High response rate.
- More expensive and time-consuming.

- ⑤ **Experimentation** : Data is collected by conducting controlled experiments and observing outcomes.

Key Points :

- Helps in establishing cause-and-effect relationship.
- Conducted under controlled conditions.
- May not reflect real-world situations always.

2. SECONDARY METHODS (Second-hand Data)

Data already collected by others for some other purpose is used.

Sources : Books, journals, research papers, reports, government publications, websites, databases, newspapers, magazines, records.

Key Points :

- Saves time, money, and effort.
- May not be exactly suited to the current research.
- Reliability and accuracy must be checked.

Note : Selection of data collection method depends on the nature of the problem, objectives, available resources, and time.

Q6. Discuss advantages and limitations of sampling techniques.

6

Sampling is the process of selecting a subset (sample) from a population to draw conclusions about the whole population.

1. ADVANTAGES OF SAMPLING

- ① Cost - Effective :
 - Sampling is much cheaper than studying the entire population.
 - Reduces expenses on data collection, processing and analysis.
- ② Time - Saving :
 - Requires less time compared to census of the whole population.
 - Enables quicker decision - making.
- ③ Feasibility :
 - Some populations are too large, infinite, or geographically dispersed; sampling makes the study feasible.
 - Destructive testing is possible (only on sample).
- ④ Greater Accuracy :
 - Better supervision and control over data collection.
 - Less chance of non-sampling errors.
 - Trained investigators can focus on sample for better quality.
- ⑤ Administrative Convenience :
 - Easier to organize, manage and process data.
 - Requires less manpower and resources.
- ⑥ Supports Detailed Study :
 - Allows more in-depth and intensive study of sample units.
 - Complex measurements can be done more carefully.
- ⑦ Statistical Efficiency :
 - Properly designed samples can give precise estimates.
 - Sampling error can be measured and controlled.

2. LIMITATIONS OF SAMPLING

- ① Sampling Error :
 - Results may differ from true population values.
 - There is always a chance of estimation error.
- ② Not Suitable for All Studies :
 - Some studies require complete enumeration (e.g., small populations).
 - Not ideal when high accuracy is needed.
- ③ Requires Skilled Personnel :
 - Improper sampling design or selection can lead to biased results.
 - Needs experienced researchers.
- ④ Non - Sampling Errors Still Possible :
 - Errors in data collection, response, or recording can still occur.
 - These can affect the results.
- ⑤ May Not Represent All Subgroups :
 - Small or rare groups may be missed in the sample.
 - Can lead to unrepresentative results.
- ⑥ Complexity in Design :
 - Some sampling techniques are complex to design and implement.
 - Requires statistical knowledge.

Note : A well-designed sample can provide reliable, valid and generalizable results while saving time, cost and effort.

7. Explain the importance of data processing in research.

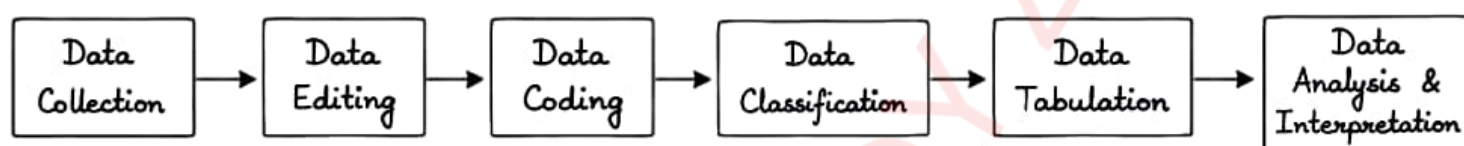
Answer:

Data processing is the systematic procedure of editing, coding, classification, tabulation and analyzing the collected data to convert it into useful information for research.

1. Meaning:

Data Processing is the transformation of raw data into meaningful and reliable information.

2. Steps in Data Processing:



3. Importance of Data Processing in Research:

- (i) Improves Accuracy: Eliminates errors and inconsistencies in raw data.
- (ii) Ensures Reliability: Provides consistent and dependable results.
- (iii) Saves Time and Cost: Reduces manual work and repetition.
- (iv) Simplifies Data: Converts large and complex data into simple and understandable form.
- (v) Facilitates Analysis: Processed data can be easily analyzed using statistical techniques.
- (vi) Supports Decision Making: Provides valid and meaningful information for better decisions.
- (vii) Enhances Presentation: Data can be presented in tables, charts and graphs for clarity.

4. Summary of Benefits:

Benefit	Explanation
Accuracy	Reduces errors and increases correctness.
Reliability	Results remain consistent across different situations.
Efficiency	Saves time and resources.
Clarity	Makes data easy to understand and interpret.
Utility	Helps in drawing valid conclusions and generalizations.

8. Explain quantitative and qualitative data analysis methods. (8)

Answer: Data analysis is the process of inspecting, cleaning, transforming and interpreting data to extract meaningful information and support research conclusions.

It is broadly classified into two types:

- (i) Quantitative Data Analysis
- (ii) Qualitative Data Analysis

1. Quantitative Data Analysis

It deals with numerical data and uses statistical and mathematical techniques to analyze and interpret data.

Objectives:

- To summarize data.
- To identify patterns and relationships.
- To test hypotheses and make generalizations.

Methods / Techniques:

- ① Descriptive Statistics: Summarizes data using measures like mean, median, mode, standard deviation, variance, range, percentages, etc.
- ② Inferential Statistics: Used to estimate population parameters and test hypotheses using techniques such as t-test, chi-square test, ANOVA, confidence intervals, etc.
- ③ Correlation & Regression: Measures relationship between variables and predicts outcomes.
- ④ Data Visualization: Represents data in forms like tables, bar charts, pie charts, histograms, line graphs, scatter plots, etc.
- ⑤ Probability Distributions: Describes the likely values of a random variable using distributions such as normal, binomial, poisson, etc.

Key Point:

Quantitative analysis converts numbers into meaningful results using statistical tools and helps in objective decision making.

2. Qualitative Data Analysis

It deals with non-numerical (categorical or textual) data and uses interpretative techniques to understand meanings, experiences and concepts.

Objectives:

- To understand meanings and context.
- To explore attitudes, opinions and experiences.
- To generate themes and patterns.

Methods / Techniques:

- ① Thematic Analysis: Identifies and analyzes patterns (themes) in data.
- ② Content Analysis: Systematic coding and classification of text or documents.
- ③ Narrative Analysis: Examines stories and personal experiences.
- ④ Grounded Theory: Develops theories based on collected data.
- ⑤ Coding: Assigning codes to responses and grouping similar codes.

Key Point:

Qualitative analysis focuses on depth, context and meaning rather than numbers and helps in understanding complex human behavior.

Q.9 Explain Paired and Unpaired t-Tests with Examples.

9

★ Introduction

A t-test is a statistical hypothesis test used to determine whether there is a significant difference between the means of two groups when the population variance is unknown and the sample size is small.

t-test compares sample means.

1. Unpaired t-Test (Independent t-Test)

Definition

An Unpaired t-Test is used when two samples are independent of each other and there is no relationship between the observations in the two groups.

Independent Samples → Unpaired t-Test

Conditions

- Two different groups
- Samples are independent
- Population variance unknown
- Data approximately normally distributed

Formula

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where,

$$S_p^2 = \frac{(n_1 - 1) S_1^2 + (n_2 - 1) S_2^2}{n_1 + n_2 - 2}$$

Where

$\bar{X}_1, \bar{X}_2$ = Sample Means

S_1^2, S_2^2 = Sample Variances

n_1, n_2 = Sample Sizes

Example

Compare the average marks of :

→ Group A → CSE Students

→ Group B → ECE Students

The two groups contain different students and are independent.

∴ Therefore, use Unpaired t-Test.

2. Paired t-Test (Dependent t-Test)

Definition

A Paired t-Test is used when observations are taken in pairs or when the same subjects are measured twice.

Related Samples → Paired t-Test

Conditions

- Same subjects tested twice
- OR
- Matched pairs of observations
- Data differences follow normal distribution

Procedure

For each pair, $d_i = X_i - Y_i$

Calculate mean difference

$$\bar{d} = \frac{\sum d_i}{n}$$

Calculate standard deviation of differences

$$S_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n - 1}}$$

Formula

$$t = \frac{\bar{d}}{S_d / \sqrt{n}} = \frac{\bar{d} \sqrt{n}}{S_d}$$

where

$\bar{d}$ = Mean Difference

S_d = Standard Deviation of Differences

n = Number of Pairs

Example

A teacher records students' marks :

Before Training → After Training

Same students are tested twice.

Since observations are related,

∴ Use Paired t-Test.

★ Difference Between Paired and Unpaired t-Test

	Paired t-Test	Unpaired t-Test
①	Dependent Samples	Independent Samples
②	Same subjects measured twice	Different subjects
③	Uses differences between pairs	Uses separate sample means
④	Related observations	Unrelated observations

Paired → Related Data

Unpaired → Independent Data

★ Conclusion

The Paired t-Test is applied when observations are related or repeated on the same subjects, whereas the Unpaired t-Test is applied when two independent groups are compared. Both tests help determine whether the difference between means is statistically significant.

★ Exam Point :

Same Subjects → Paired t-Test

Different Subjects → Unpaired t-Test

10. Explain Type-I and Type-II errors with examples.

Type-I and Type-II Errors

In discrete mathematics and hypothesis testing, errors can occur when we make wrong decisions while testing a statement (hypothesis).

① Type-I Error (False Positive)

- It occurs when we reject a true hypothesis.
- We conclude that a statement is false, but in reality it is true.
- Denoted by α (alpha)
- Called level of significance.

② Type-II Error (False Negative)

- It occurs when we fail to reject a false hypothesis.
- We conclude that a statement is true, but in reality it is false.
- Denoted by β (beta)
- Called Type-II error probability.

Relationship between hypothesis and errors

Decision	Actual Situation	
	H_0 is True	H_0 is False
Reject H_0	Type-I Error (Probability = α)	Correct Decision (Probability = $1 - \beta$)
Do not Reject H_0	Correct Decision (Probability = $1 - \alpha$)	Type-II Error (Probability = β)

Important Notes :

- Smaller α reduces Type-I error but may increase Type-II error.
- Smaller β increases the power of the test (Power = $1 - \beta$).
- There is a trade-off between α and β .

Summary :

Type-I Error $\rightarrow$ Reject a true hypothesis (False Positive) $\rightarrow$ Probability = α
 Type-II Error $\rightarrow$ Do not reject a false hypothesis (False Negative) $\rightarrow$
 Probability = β

Q11. Discuss one-way and two-way ANOVA.

11

ANOVA (Analysis of Variance) is used to compare the means of three or more groups to determine whether at least one group mean is significantly different from the others.

1. ONE-WAY ANOVA (One independent variable or factor)

Purpose : To compare the means of three or more groups based on one factor.

When to use : When there is only one independent variable (factor) affecting the dependent variable.

Hypotheses :
 H_0 : All group means are equal ($\mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$)
 H_1 : At least one group mean is different.

Test Statistic : $F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$ **Where,** $MS_{\text{between}} = \frac{SS_{\text{between}}}{df_{\text{between}}}$

- SS_{between} = Sum of Squares between groups
 - SS_{within} = Sum of Squares within groups (error)
 - $df_{\text{between}} = k - 1$ (k = number of groups)
 - $df_{\text{within}} = N - k$ (N = total number of observations)
- $MS_{\text{within}} = \frac{SS_{\text{within}}}{df_{\text{within}}}$

Decision Rule : If $F_{\text{calculated}} > F_{\text{critical}}$ (from F-table at α level), Reject H_0 . Otherwise, fail to reject H_0 .

Conclusion : If H_0 is rejected, conclude that at least one group mean is significantly different.

2. TWO-WAY ANOVA (Two independent variables or factors)

Purpose : To study the effect of two factors on a dependent variable and also check if there is an interaction between the factors.

When to use : When there are two independent variables (factors) that may influence the dependent variable.

Hypotheses :

- **Factor A (rows):** H_0 : No effect of factor A
 H_1 : Factor A has an effect
- **Factor B (columns):** H_0 : No effect of factor B
 H_1 : Factor B has an effect
- **Interaction (A×B):** H_0 : No interaction between A and B
 H_1 : There is an interaction between A and B

Test Statistic : For each source (A, B, Interaction, Error),

$$F = \frac{MS_{\text{source}}}{MS_{\text{error}}} \quad \text{Where, } MS_{\text{source}} = \frac{SS_{\text{source}}}{df_{\text{source}}}$$

$$MS_{\text{error}} = \frac{SS_{\text{error}}}{df_{\text{error}}}$$

Sources of Variation	SS	df	MS	F
Factor A (Rows)	SS_A	$a - 1$	MS_A	MS_A / MS_{error}
Factor B (Columns)	SS_B	$b - 1$	MS_B	MS_B / MS_{error}
Interaction (A×B)	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	$MS_{AB} / MS_{\text{error}}$
Error (Within)	SS_E	$ab(n-1)$	MS_E	-
Total	SS_T	$abn - 1$	-	-

Where,

a = number of levels of Factor A
 b = number of levels of Factor B
 n = number of observations per cell
 N = total observations, $N = a \times b \times n$

Decision Rule : For each F (A, B, A×B), if $F_{\text{calculated}} > F_{\text{critical}}$, reject the respective H_0 .

Conclusion : Determine which factor(s) (A, B) and/or interaction (A×B) have a significant effect on the dependent variable.

Key Differences :	One-Way ANOVA	Two-Way ANOVA
	• One independent variable (factor). • Compares means of 3 or more groups. • No interaction effect. • Simpler model.	• Two independent variables (factors). • Analyzes main effects of both factors. • Also studies interaction effect (A×B). • More complex model.

Note : ANOVA assumes normality, independence, and homogeneity of variances. Post-hoc tests (e.g., Tukey, Bonferroni) are used after ANOVA if F is significant to find which means are different.

Q12. Compare t-test and ANOVA.

Both t-test and ANOVA are inferential statistical tests used to compare means and determine whether the observed differences are statistically significant.

1. t-TEST (Student's t-test)

Purpose : Used to compare the means of two groups.

- Types :**
- One-sample t-test : Compare sample mean with a known population mean.
 - Independent (two-sample) t-test : Compare means of two independent groups.
 - Paired (dependent) t-test : Compare means of two related/paired samples.

Test Statistic :

• One-sample t-test : $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

$df = n - 1$

- Independent two-sample t-test (equal variances) :

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{where, } s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$$

$df = n_1 + n_2 - 2$

- Paired t-test :

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

$df = n - 1$

Where,

$\bar{x}, \bar{x}_1, \bar{x}_2$ = Sample mean(s)

μ = Population mean

s = Sample standard deviation

n, n_1, n_2 = Sample size(s)

$\bar{d}$ = Mean of differences

s_d = Standard deviation of differences

Characteristics :

- Compares only two means at a time.
- Assumes data is approximately normally distributed.
- Used for small to moderate sample sizes.
- Sensitive to outliers.

2. ANOVA (Analysis of Variance)

Purpose : Used to compare the means of three or more groups.

- Types :**
- One-way ANOVA : One independent variable (factor).
 - Two-way ANOVA : Two independent variables (factors).
 - Multi-way ANOVA : More than two independent variables.

Test Statistic (One-way ANOVA) :

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

where,

$$MS_{\text{between}} = \frac{SS_{\text{between}}}{df_{\text{between}}}$$

$$MS_{\text{within}} = \frac{SS_{\text{within}}}{df_{\text{within}}}$$

Where,

SS = Sum of Squares

df = Degrees of Freedom

MS = Mean Square

Characteristics :

- Compares three or more means simultaneously.
- Tests whether at least one group mean is different.
- Assumes normality, independence, and homogeneity of variances.
- More robust to outliers compared to multiple t-tests.
- If F-test is significant, post-hoc tests (e.g., Tukey, Bonferroni) are used to find which means differ.

3. Key Differences

Basis	t-test	ANOVA
No. of Groups Compared	Two groups only	Three or more groups
Test Statistic	t-distribution	F-distribution
Purpose	Compare means of 2 groups	Compare means of 3 or more groups
Types	One-sample, Independent, Paired	One-way, Two-way, Multi-way
Assumptions	Normality, Independence	Normality, Independence, Homogeneity of variances
Post-hoc Test	Not required	Required if F-test significant
Sensitivity to Outliers	High	Relatively less

Note :

- Use t-test when comparing two means.
- Use ANOVA when comparing three or more means.



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<https://whatsapp.com/channel/0029VbBFL7g0lwgnEdhCNT1T>

YouTube

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<https://debkantastudyzone.edgeone.app/>

WhatsApp Group 2023-27

<https://chat.whatsapp.com/IUvBRFPUtigAOdtleZJ6Bn>

WhatsApp Group 2024-28

<https://chat.whatsapp.com/DTgDyoXsMabDfuMTkJ8k2I>

WhatsApp Group 2025-29

<https://chat.whatsapp.com/BVR1vA0Hgzt6oYic59EQqn>

GATE & Placement

<https://chat.whatsapp.com/EFnokgMa1UOEPLcqw4Wm8u>

ECE 2024-28

<https://chat.whatsapp.com/ELWjlZcYEMGK5ktfxmjLt4>

ECE 2023-27

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